Reads over Atom Spaces: A Mathematical View of Context, Retrieval, and Parameter Routing in Language Models

Farshid Mossaiby[a,*]

[a]*Department of Civil Engineering, University of Isfahan, 81744-73441, Isfahan, Iran*

Keywords: Language models, Atom spaces, Vector retrieval, Expert routing, Model editing

---

Research Highlights

- One read operator over atom spaces covers attention, retrieval, and MoE routing
- A null atom yields abstention; the retrieval threshold is its zero-temperature limit
- An anti-atom is deletion plus a null mass: it abstains where deletion falls through
- A query-side margin certifies a top-1 radius at insertion with no Lipschitz premise
- A rank-margin gate lifts CounterFact paraphrase activation to 71% at 7% false hits

*Corresponding author.
Email address: mossaiby@eng.ui.ac.ir (Farshid Mossaiby)